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Module 8 Introduction

Lesson 1: Setting Up and Interpreting Mixture Models

Lesson 2: Practical Applications and Computational Challenges

- **Reading:** Practical Applications and Computational Challenges Readings
  3h
- **Video:** Applications of Mixture Models Pt. 1
  9 min
- **Video:** Applications of Mixture Models Pt. 2
  5 min
- **Video:** Applications of Mixture Models Pt. 3
  7 min
- **Practice Assignment:** Practical Applications and Computational Challenges Quiz
  Submitted
- **Lab:** Module 8 - Lesson 2 - RStudio Lab
  1h

Module 8 Summative Assessment

Module 8 Summary

Practical Applications and Computational Challenges Quiz

Review Learning Objectives

Assignment details

Submitted
Jan 11, 12:19 AM IST

Attempts
Unlimited

Retry

Your grade
To pass you need at least 80%. We keep your highest score.

100%

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Next item →

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Instructions

1. What is an example of a practical problem where mixture distributions arise? **1 / 1 point**

- ○ Determining the average height in a population
- ◉ Analyzing the reaction times of schizophrenics on an attentional task
- ○ Calculating the total sales of a product
- ○ Measuring the weight of a single group of items

Correct
Correct! This is an example where mixture distributions can be used.

2. What does the term 'posterior predictive distribution' refer to in Bayesian mixture models? **1 / 1 point**

- ○ The distribution of observed data
- ○ The prior distribution of parameters
- ○ The likelihood function
- ◉ The distribution of future observations based on the posterior distribution of parameters

Correct
Correct! The posterior predictive distribution is the distribution of future observations based on the posterior distribution of parameters.

3. What is a common problem when choosing the number of mixture components, H, in a finite mixture model? **1 / 1 point**

- ○ It simplifies the computation
- ○ It often leads to overfitting
- ◉ There is uncertainty about the correct number of components
- ○ It always reduces model flexibility

Correct
Correct! There is often uncertainty about the appropriate number of mixture components to include.